

Design and Analysis of Algorithms

Date: Nov 25, 2025

Course Instructor(s)

Dr. Fahad Sherwani

Quiz - 2

Total Time: 45 Min

Total Marks: 2.5

Total Questions: 3

**** Plagiarism of any kind is highly discouraged and will result in Zero marks ****

Roll No

Section

Student Signature

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Attempt all the questions.

CLO #4: To construct and analyze real world problems solutions using different algorithms design techniques i.e. Brute Force, Divide and Conquer, Dynamic Programming, Greedy Algorithms.

Q1: A power distribution company is planning to install underground electrical cables to connect eight rural locations. Laying a cable between any two locations has a specific construction cost. The planning team provides the following information:

- Each location is represented as a node.
- A cable between two locations is represented as an undirected edge with a cost.

The company wants to connect all locations while:

- Keeping the total installation cost as low as possible, and
- Ensuring no redundant loops or cycles exist in the final network.

The possible cable connections and their associated installation costs are:

(1–2, 4), (1–3, 6), (2–3, 2), (2–4, 5), (3–4, 1), (3–5, 3), (4–5, 7)

- a) Explain why a greedy approach is appropriate for minimizing the total wiring cost while also avoiding unnecessary cycles.
- b) Using the data above, design an algorithm that determines the set of cable connections the company should install to achieve the lowest possible total construction cost under the stated constraints.

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Q2: A delivery company operates inside a large warehouse containing hundreds of shelves and storage racks. Workers move between racks using narrow walkable passages. This warehouse is modeled as an unweighted undirected graph, where:

- Each node represents a rack.
- Each edge represents a walkable passage between racks.

A worker starts at rack S, and wants to know:

- Which racks are reachable by exploring the warehouse systematically, starting from S.
 - The minimum number of steps required to reach any rack in the warehouse.
- a) Design an algorithm to find the shortest number of steps from S to every other rack.
b) For the graph below, show the results using your algorithm and the best exploration order from Edges: (1–2), (1–3), (2–4), (2–5), (3–6), (5–6), (4–7) Let S = 1.

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Q3: A university is designing an academic progression plan for a new degree program. Each course may require students to complete one or more prerequisite courses before enrolling. The curriculum committee provides the following prerequisite structure:

<u>Course</u>	<u>Prerequisite(s)</u>
C1	—
C2	C1
C3	C1
C4	C2, C3
C5	C2
C6	C4, C5
C7	C4
C8	C6, C7

- a) Draw a directed graph that represents the prerequisites, where each course is a node and each prerequisite relation is a directed edge.
b) Design an algorithm to generate a valid course-taking order such that all prerequisite requirements are satisfied and provide one valid order in which a student may complete all eight courses.